

Time spent: 0 sec

Subject:
Biology

QID: 700920

System:
Digestion and Excretion

Renal reabsorption of glucose is facilitated by sodium-glucose linked cotransporters (SGLTs), which are secondary active transport proteins located in the apical membrane of the proximal tubular cells in the kidney. Glucose transported out of the proximal tubule via SGLTs is returned to circulation through the peritubular capillaries. SGLT activity determines the transport maximum of glucose T_m , the maximum rate at which the kidney can reabsorb glucose.

Excretion of glucose in the urine (glycosuria) occurs when the glomerular filtration rate (GFR) of glucose exceeds the T_m . Glycosuria is a symptom of diabetes mellitus, a disease marked by hyperglycemia (elevated serum glucose). Uncontrolled hyperglycemia can damage the nerves that innervate urinary tract musculature, resulting in urinary retention (inability to empty the bladder completely) or incontinence (involuntary urine leakage).

Canagliflozin is an SGLT inhibitor used in the treatment of diabetes. The effect of canagliflozin on renal glucose reabsorption was tested in a glucose infusion experiment in patients with type 2 diabetes mellitus (T2DM). First, researchers measured the participants' fasting serum glucose levels (average ~80 mg/dL) and then began a continuous intravenous infusion of a glucose solution. The rate of glucose infusion was adjusted such that plasma glucose concentration steadily rose to 300 mg/dL. Plasma and urine glucose concentrations were measured at 30-minute intervals. The experiment was later repeated in the same patients following administration of canagliflozin (100 mg once a day for 7 days). The results of both trials are shown in Figure 1.

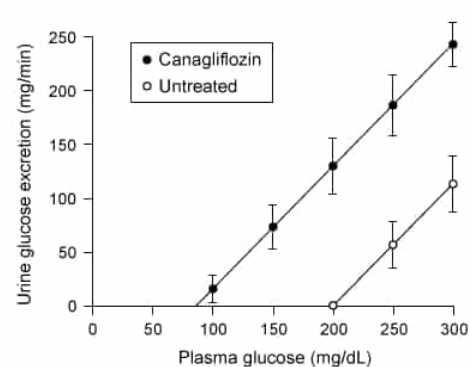


Figure 1 Mean urine glucose excretion vs. blood plasma concentration in patients with T2DM, with and without canagliflozin treatment

Researchers then estimated the new T_m of glucose following canagliflozin treatment (Figure 2). Canagliflozin was found to also lower fasting plasma glucose in individuals with T2DM. In addition to the blood pressure reduction that typically accompanies canagliflozin therapy, common adverse effects included weight loss, increased urinary frequency, and excessive thirst.

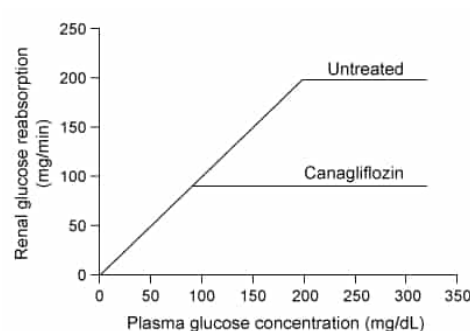


Figure 2 Mean rate of renal glucose reabsorption vs. plasma glucose concentration in patients with T2DM, with and without canagliflozin treatment

Researchers hypothesized that administration of canagliflozin reduces blood pressure in patients with T2DM by increasing urine output and reducing total blood volume. A decrease in which of the following would also result in lowered blood pressure?

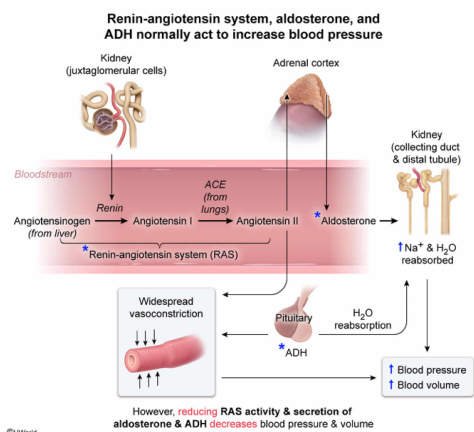
- I. Activation of the renin-angiotensin system
- II. Production of aldosterone
- III. Secretion of antidiuretic hormone

- ☐ A. II only [13%]
☐ B. III only [15%]
☐ C. I and III only [13%]
☒ D. I, II, and III [58%]

Correct

57% answered correctly

Explanation:



Blood pressure (BP), the force blood exerts on blood vessel walls, decreases when blood vessels **dilate** (become wider) or when **blood volume** decreases. Hormones such as angiotensin II, aldosterone, and

antidiuretic hormone (ADH; vasopressin) regulate BP by **modulating** reabsorption of water and salts by the kidney.

The **renin-angiotensin system (RAS)** is a multiorgan molecular cascade activated when BP (or blood volume) falls. A drop in BP causes the juxtaglomerular cells in the kidney to release renin, an enzyme that cleaves the plasma protein angiotensinogen to form angiotensin I. Angiotensin-converting enzyme then cleaves angiotensin I to form angiotensin II. Angiotensin II ultimately raises BP by inducing both the **release of aldosterone** from the adrenal cortex (increasing BP by increasing blood volume through water retention) and the **constriction of arterioles** (increasing BP without changing blood volume).

Aldosterone is released in response to RAS activation or to an increased serum level of K⁺. Aldosterone acts on the distal tubules and collecting ducts of nephrons to promote the **reabsorption of Na⁺** and the **secretion of K⁺**. Increased reabsorption of Na⁺ increases the osmolarity, or solute concentration, of the renal interstitial fluid. Elevated osmolarity promotes **water reabsorption**, which ultimately causes blood volume and BP to increase.

ADH is released by the posterior **pituitary** when BP falls or when blood osmolarity rises. ADH promotes **water reabsorption** by increasing the **permeability** of the distal tubule and collecting duct to water. It also induces **vasoconstriction**, the narrowing of blood vessels. Both of these effects increase BP.

Angiotensin II of the RAS pathway, aldosterone, and ADH are hormones that cause BP to *increase*. Accordingly, **a decrease in RAS activation, ADH secretion, or aldosterone production** would **cause BP to fall (Numbers I, II, and III)**.

Educational objective:

Angiotensin II, antidiuretic hormone, and aldosterone all regulate blood pressure. The effect of these hormones on blood pressure can be explained wholly or in part by the modulation of water and salt reabsorption in the kidney.

Time spent:0 sec

Subject:

Biology

QID:400927

System:

Digestion and Excretion

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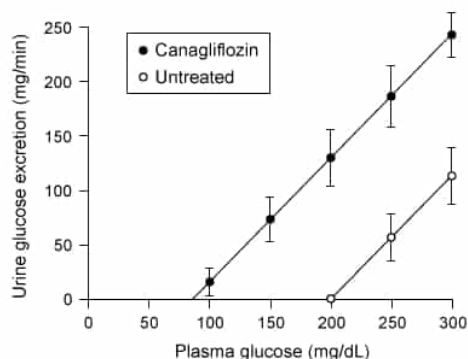


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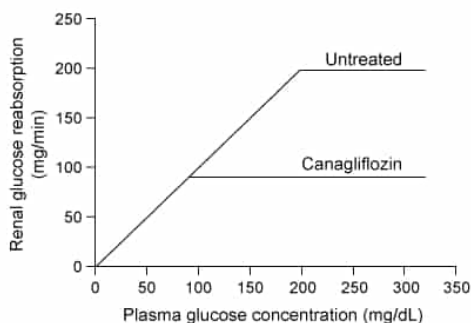


Figure 2 Mean rate of renal glucose reabsorption vs. plasma glucose concentration in patients with T2DM, with and without canagliflozin treatment

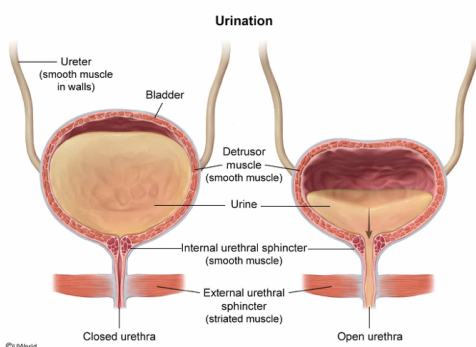
Urinary retention in certain patients with poorly controlled diabetes mellitus is most likely the result of nerve damage that reduces the ability of smooth muscle to contract in which of the following structures?

- ✓ ☒ A. Bladder [48%]
☐ B. Urethral sphincters [43%]
☐ C. Kidney efferent arterioles [2%]
☐ D. Ureter [5%]

Correct

49% answered correctly

Explanation:



After being formed in the nephron, urine is **funneled** into the **ureters** (the tubes connecting the kidneys to the bladder) for delivery to the bladder. Urine accumulates in the bladder until it exits the body via the urethra in a process called **urination**.

Urination is controlled by musculature (both smooth and striated) lining the urinary tract. During urine collection, the **detrusor muscle** (the layer of *smooth muscle* lining the bladder) is relaxed. At the same time, the **internal urethral sphincter** (IUS, a ring of smooth muscle surrounding the urethra) and the **external urethral sphincter** (EUS, a ring of striated muscle downstream of the IUS) are contracted to prevent urine flow down the urethra. The detrusor and IUS are under involuntary control; the EUS is under voluntary control.

During urination, activity within stretch receptors in the bladder leads to:

- Contraction of the detrusor muscle, which pushes urine out of the bladder and into the urethra
- Relaxation of the IUS, which opens the urethra and allows urine to pass

Nerve damage that *reduces* the contraction of the *detrusor muscle* of the *bladder* impairs emptying of the bladder, leading to *urinary retention*. Therefore, the **bladder** is the only structure of those listed that might be responsible for urinary retention because the **smooth muscle** of its **detrusor muscle** has **reduced ability to contract** if it sustains nerve damage.

(Choice B) Urine can pass through the urethra and out of the body when the IUS and EUS are relaxed. When contracted, both the IUS and EUS block urine flow through the urethra. Therefore, a *reduced* ability to contract these muscles due to nerve damage would likely increase, *not* impair, urine flow.

(Choice C) Diminished contraction of the smooth muscles (eg, of the efferent arteriole carrying blood out of the glomerulus) in the kidney due to nerve damage may alter the rate of urine *production* but would *not* affect the process of urination.

(Choice D) Decreased contraction of smooth muscles in the ureter due to nerve damage may affect *urine collection* in the bladder. However, the ability to urinate, which is under control of the detrusor muscle, IUS, and EUS, would *not* be affected.

Educational objective:

Urine flows from the kidney to the bladder through the ureter. Once the bladder is full, urine exits the body via the urethra through the process of urination. Urination is controlled by the detrusor muscle (smooth muscle lining the bladder), the internal urethral sphincter, and the external urethral sphincter.

Time spent:0 sec

Subject:
Biology

QID:400928

System:
Digestion and Excretion

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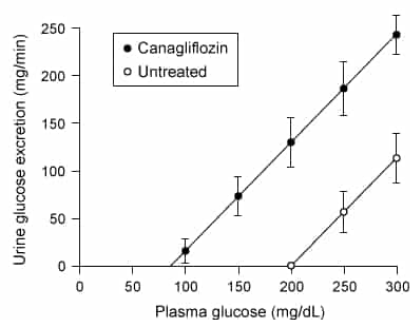


Figure 1 Mean urine glucose excretion vs. blood plasma concentration in patients with T2DM, with and without canagliflozin treatment

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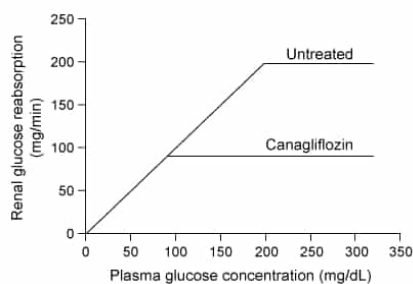


Figure 2 Mean rate of renal glucose reabsorption vs. plasma glucose concentration in patients with T2DM, with and without canagliflozin treatment

Administration of canagliflozin increases urinary output most directly by increasing the:

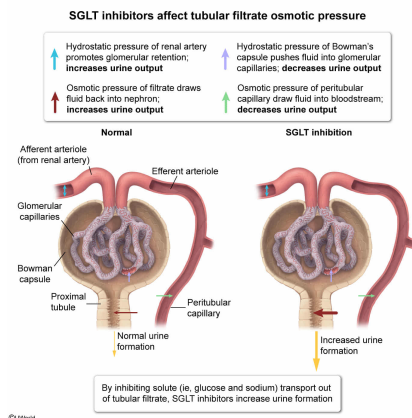
- ✓ ☒ A. osmotic pressure of the tubular filtrate. [52%]
- ☐ B. osmotic pressure of the peritubular capillaries. [14%]

- ☐ C. hydrostatic pressure of the renal artery. [12%]
- ☐ D. hydrostatic pressure of Bowman's space. [21%]

Correct

51% answered correctly

Explanation:



Hydrostatic pressure and osmotic pressure **regulate fluid movement** between capillaries and the tissues that surround them. Hydrostatic pressure is the force exerted by a liquid on the walls of its container. Accordingly, the **hydrostatic pressure of blood**, or blood pressure (BP), is the force exerted by blood on vessel walls. This pressure **pushes fluid** through capillary wall pores and **into the interstitial fluid** (the extracellular fluid surrounding cells of a tissue). In contrast, **osmotic pressure** is the force drawing water into a solution by **osmosis**, the passive movement of water molecules from an area of low to an area of high solute concentration across a semipermeable membrane. Osmotic pressure increases as solute concentration (**osmolarity**) increases.

The kidneys contribute to BP regulation by increasing water reabsorption when BP is low (greater blood volume facilitates higher BP) and decreasing water reabsorption when BP is high. In addition, the kidneys respond to high blood osmolarity by increasing water reabsorption and decreasing solute reabsorption. Conversely, when blood osmolarity is low, the kidneys react by decreasing water reabsorption and increasing solute reabsorption.

The passage states that SGLTs cotransport glucose and sodium out of the proximal tubule so they can be returned to the circulation through the peritubular capillaries. Therefore, SGLT *inhibitors* (eg, canagliflozin) most likely *prevent* glucose and sodium reabsorption from the tubule. This leads to *increased* solute concentration within the tubule, thereby **increasing** the **osmotic pressure** of the **tubular filtrate**. Consequently, less water is reabsorbed, and *more* water is drawn into the tubule, **increasing urine output**.

(Choice B) Increasing the osmotic pressure of blood in the peritubular capillaries would *increase* water retention by osmosis and *decrease* urinary output.

(Choice C) According to the passage, SGLT inhibitors like canagliflozin typically decrease (*not* increase) BP. If decreased hydrostatic pressure in the renal artery also reduced glomerular capillary pressure, the decreased pressure would *reduce* the amount of fluid being filtered into Bowman's capsule from the glomerular capillaries, thereby decreasing (*not increasing*) overall urine output.

(Choice D) Hydrostatic pressure within the glomerulus pushes fluid into Bowman's space. This movement is opposed by the hydrostatic pressure in Bowman's space. Therefore, increasing the hydrostatic pressure in Bowman's space would impede fluid from entering the nephron and would *decrease* urinary output.

Educational objective:

The kidneys regulate blood osmolarity and, in turn, blood pressure by selectively modulating the reabsorption and secretion of water and solutes.

Time spent: 0 sec
Subject:
Biology

QID: 400929
System:
Digestion and Excretion

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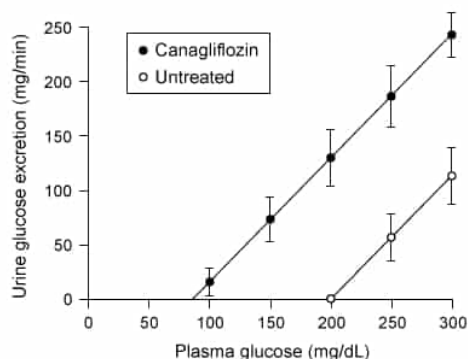


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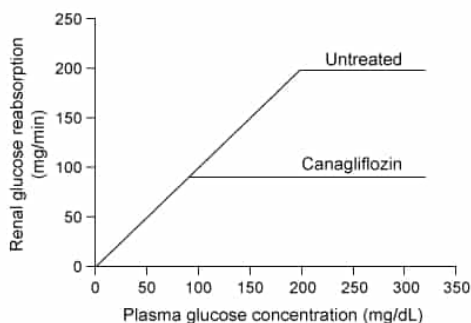


Figure 2 Mean rate of renal glucose reabsorption vs. plasma glucose concentration in patients with T2DM, with and without canagliflozin treatment

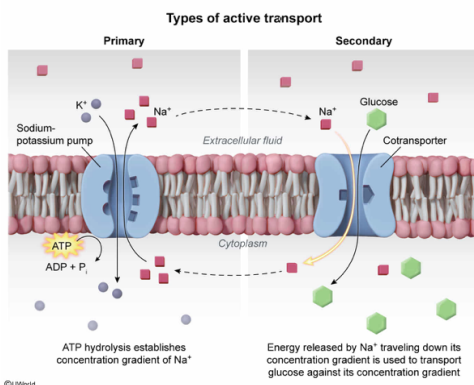
SGLTs transport glucose against its concentration gradient across the luminal membrane of the proximal tubule. The energy to drive this form of glucose transport is most likely directly provided by:

- ✓ ☒ A. an electrochemical Na^+ gradient. [48%]
☐ B. an action potential. [1%]
☐ C. ATP hydrolysis. [37%]
☐ D. simple diffusion of Na^+ ions. [12%]

Correct

48% answered correctly

Explanation:



Substances can be transported across membranes through **various processes**. Passive transport is the movement of a substance across a cell membrane *along* its concentration gradient (ie, from an area of high concentration to an area of low concentration); it *does not* require energy. In contrast, active transport is the movement of a substance across a cell membrane *against* its concentration gradient (ie, from an area of low concentration to an area of high concentration). Active transport *does* require energy.

There are two types of active transport:

- **Primary** (direct) **active transport** uses the energy released by ATP hydrolysis (performed by a transmembrane ATPase).
- **Secondary** (indirect) **active transport** is a coupled transport process that uses the energy released by the movement of one substance *along* its concentration gradient (passive transport) to move another substance *against* its concentration gradient. Essentially, the potential energy stored in the concentration gradient of one molecule is used for the transport of another molecule.

According to the passage, **SGLTs** are **secondary active transport proteins**. This means that SGLTs use the energy released by the movement of Na^+ *along* its concentration gradient to move glucose *against* its concentration gradient (back *into* the body). For charged particles such as Na^+ , this movement occurs in reference to an **electrochemical gradient**, which incorporates both the **electrical** (potential) and **chemical** (concentration) gradients of the molecule. Accordingly, the **movement of Na^+ along its electrochemical gradient** enables glucose reabsorption.

(Choice B) An **action potential** is a brief reversal of cell membrane polarization that involves the movement of ions through membrane channels *along* their concentration gradients. Action potentials are not involved in the transport of substances (eg, glucose) *against* their concentration gradients.

(Choice C) By establishing the electrochemical gradient of Na^+ , ATP hydrolysis by the Na^+/K^+ pump *indirectly* powers the secondary active transport of glucose. However, glucose reabsorption is *directly* facilitated by the potential energy that is released when Na^+ flows along its concentration gradient.

(Choice D) Because it is charged, Na^+ cannot cross directly through the lipid bilayer via *simple diffusion*. Instead, it crosses the membrane via *facilitated diffusion* (ie, with the help of membrane proteins).

Educational objective:

Primary active transport uses the energy from ATP hydrolysis to move a substance against its concentration gradient. Secondary active transport uses the energy released by the passive transport of another molecule.

Time spent:0 sec

Subject:
Biology

QID:400930

System:
Digestion and Excretion

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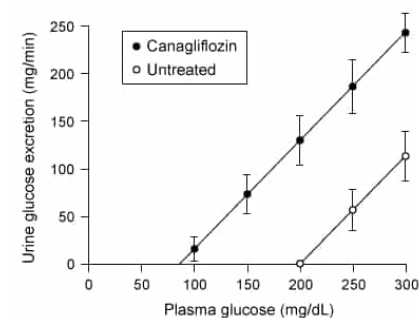


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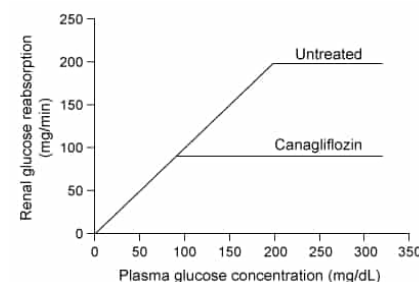


Figure 2 Mean rate of renal glucose reabsorption vs. plasma glucose concentration in patients with T2DM, with and without canagliflozin treatment

In individuals with glycosuria, glucose that is not reabsorbed in the proximal tubule flows into which of the following nephron structures?

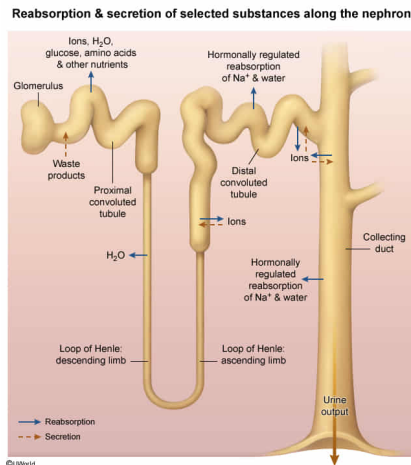
- ☐ A. The distal tubule [7%]

- ☐ B. The collecting duct [12%]
- ✓ ☒ C. The loop of Henle [74%]
- ☐ D. The Bowman capsule [5%]

Correct

74% answered correctly

Explanation:



The **nephron** is the functional unit of the **kidney**. It consists of a **glomerulus** (a bed of capillaries that receives and filters blood from the renal arteries) and a **long tubule** where filtered fluid is concentrated into urine. During urine formation, the nephron facilitates the removal (secretion) of some solutes from the blood into filtrate (fluid filtered into the nephron) as well as the **reabsorption** of useful solutes back into the blood from the filtrate.

Hydrostatic and osmotic forces **determine** the net flow of fluid from the blood through the porous glomerular capillaries into the Bowman capsule, a cup-shaped extension of the renal tubule that surrounds the glomerulus (**Choice D**). This filtrate within the capsule contains small solutes such as glucose, ions, and urea but *lacks* blood cells and proteins, which remain in circulation. The filtrate passes from the Bowman capsule into a long tubule that is composed successively of the following segments:

1. **Proximal tubule:** In this segment, important nutrients (eg, amino acids, vitamins, salts, glucose, water) are reabsorbed from the nephron and incorporated back into the blood through the peritubular capillaries that surround the tubule. Waste products not filtered by the Bowman capsule are actively secreted from the peritubular capillaries into the nephron.
2. **The loop of Henle:** This segment is a hairpin structure composed of two limbs. The *descending limb* extends down into the relatively salty **medulla** (inner portion of the kidney) and is permeable to water, allowing the passive reabsorption of water via osmosis. The *ascending limb* transports filtrate out of the medulla and back to the cortex (outer portion of the kidney). This limb is permeable to salt but impermeable to water; this leads to active reabsorption of ions only, which concentrates the medullary interstitial fluid (maintains its saltiness) and prevents its dilution.
3. **Distal tubule (Choice A):** Antidiuretic hormone (ADH; vasopressin) and aldosterone promote the reabsorption of water in this segment of the tubule. Additional waste products are secreted into the nephron from the peritubular capillaries.
4. **Collecting duct (Choice B):** ADH and aldosterone act on the collecting duct in the same way they act on the distal tubule. The now-concentrated urine is emptied into the ureters for excretion.

Educational objective:

Fluid that is filtered by the kidney first enters Bowman's capsule before passing into a long tubule where water is reabsorbed and solutes are selectively reabsorbed or secreted. Waste and metabolites are excreted as urine.

Time spent: 0 sec
Subject:
Biology

QID: 400931
System:
Digestion and Excretion